High-Dimensional Arrhythmia Classification from Raw ECG Time Series

Data understanding, preprocessing, model tuning, evaluation, and RMT diagnostics

ECG RMT Analysis

May 2026

Task Definition

Goal

Build a reproducible binary ECG screening pipeline from PTB-XL raw WFDB waveforms and metadata, then explain how fitted models behave in the high-dimensional feature regime.

- Input: ten-second, twelve-lead ECG waveform records plus patient and acquisition metadata.
- Target: normal versus diagnostic abnormality from SCP diagnostic superclasses.
- Data split: official PTB-XL folds; train, validation, and held-out test are kept separate.
- Output: trained model comparison, final metrics, paper, presentation, and MLflow logs.

EDA: Dataset and Label Construction

PTB-XL inputs

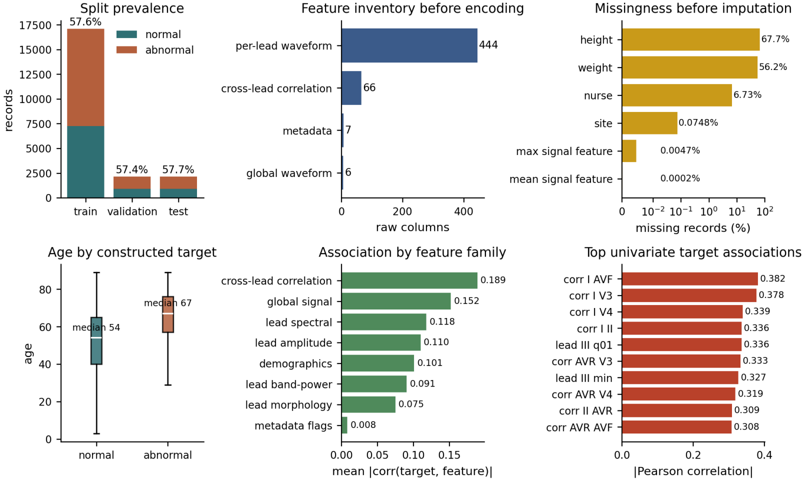
- 21,799 metadata rows.
- WFDB records at 100 Hz and 500 Hz.
- Twelve simultaneous ECG leads.
- SCP-ECG code dictionary for diagnostic mapping.

Constructed supervised table

Quantity	Count
Labeled records	21,388
Normal	9,069
Diagnostic abnormality	12,319
Unlabeled or unmapped	411

The binary target is a screening abstraction: a record is positive if at least one mapped diagnostic superclass is not `NORM`.

EDA: What We Found Before Modeling



Rows: 21,388; raw feature map: 523 columns; preprocessed design matrix: 594 columns after one-hot expansion.

EDA: Key Data Characteristics

Class and fold structure

- Train: 17,084 records.
- Validation: 2,146 records.
- Test: 2,158 records.
- Abnormal class prevalence is near 58% in each split.

Quality observations

- Height and weight are highly incomplete, so median imputation is required.
- Waveform-derived features are almost complete after decoding.
- Age differs between target groups, so demographics are retained but regularized.
- Cross-lead correlations show strong univariate association with the target.

Preprocessing Pipeline

WFDB decoding → SCP code mapping → ECG feature extraction → official fold split → train-only preprocessing → model and RMT analysis

Leakage control

Imputation, scaling, and one-hot expansion are fit on training records only and then applied to validation/test records through the same pipeline object.

Preprocessing and Feature Handling

Numeric features	Categorical metadata	Feature dimensions
median imputation followed by standard scaling for linear models and covariance analysis	most-frequent imputation and one-hot encoding with unknown categories ignored at test time	523 raw modeling columns become 594 preprocessed columns after encoding

$$z_i = \phi(\text{waveform}_i, \text{metadata}_i) \in \mathbb{R}^{523}, \qquad x_i = g(z_i) \in \mathbb{R}^{594}.$$

Feature Engineering Choices

Feature group	Reason for inclusion
Lead amplitude statistics	Capture waveform scale, spread, asymmetry, and robust quantiles.
Derivative and energy features	Represent local morphology and signal roughness without hand-labeled beats.
FFT and band-power features	Add spectral information for rhythm and noise-sensitive structure.
Cross-lead correlations	Encode multi-lead coordination and axis-like relationships.
Metadata	Retain clinically relevant acquisition and demographic context.

Design choice

The project uses interpretable engineered features instead of a deep ECG network so that preprocessing, feature families, and RMT spectra are directly inspectable.

Model Families and Why They Were Chosen

Model	Purpose in the comparison
Logistic regression	Regularized linear baseline for high-dimensional tabular features.
SGD linear	Fast linear margin model with tunable L_1/L_2-style regularization.
Random forest	Bagged nonlinear ensemble with variance reduction.
Extra trees	More randomized tree ensemble to test sensitivity to split randomness.
Histogram gradient boosting	Strong CPU boosting baseline with explicit L_2 regularization.
XGBoost	Regularized boosted trees with L_1, L_2, subsampling, and column sampling.

Training and Tuning Protocol

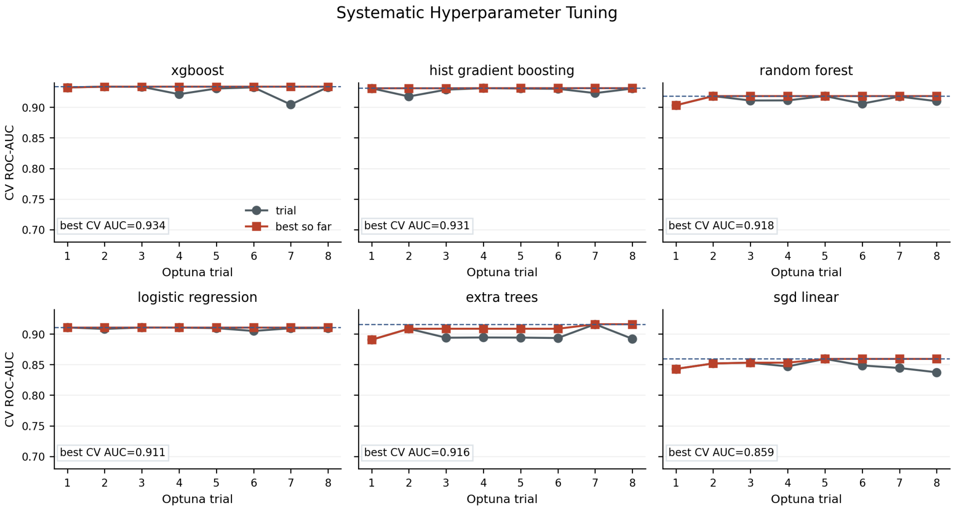
Methodology

- Train fold is used for fitting and cross-validation.
- Optuna TPE search optimizes CV ROC-AUC.
- Validation fold checks model-family selection.
- Test fold is held out for final reported metrics.

Tuned hyperparameters

- Linear: C, α, penalty, L_1 ratio.
- Forests: trees, depth, leaf size, feature sampling.
- Boosting: learning rate, iterations, leaves, L_2.
- XGBoost: depth, trees, sampling, L_1, L_2.

Systematic Hyperparameter Tuning



Evaluation Metrics

Why multiple metrics?

The target is moderately imbalanced, so accuracy alone is not enough. We report ROC-AUC, F1, balanced accuracy, precision, recall, and confusion matrices.

Metric	Interpretation
ROC-AUC	Ranking quality across thresholds.
F1	Precision-recall balance at the operating threshold.
Balanced accuracy	Average of sensitivity and specificity.
Confusion matrix	Error type: missed abnormal records versus false alarms.

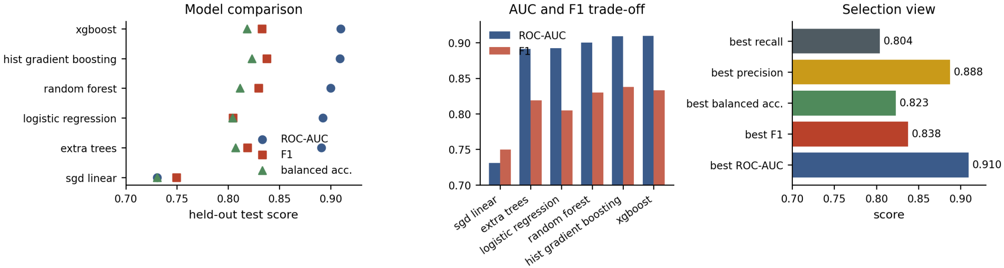
Final Held-Out Test Metrics

Model	Acc.	Bal. acc.	Prec.	Recall	F1	ROC-AUC
Logistic regression	0.7938	0.8044	0.8877	0.7360	0.8047	0.8924
SGD linear	0.7271	0.7307	0.7973	0.7071	0.7495	0.7307
Random forest	0.8100	0.8115	0.8597	0.8018	0.8297	0.9001
Extra trees	0.8021	0.8072	0.8686	0.7745	0.8188	0.8909
Hist. gradient boosting	0.8202	0.8231	0.8743	0.8042	0.8378	0.9089
XGBoost	0.8151	0.8183	0.8712	0.7978	0.8328	0.9097

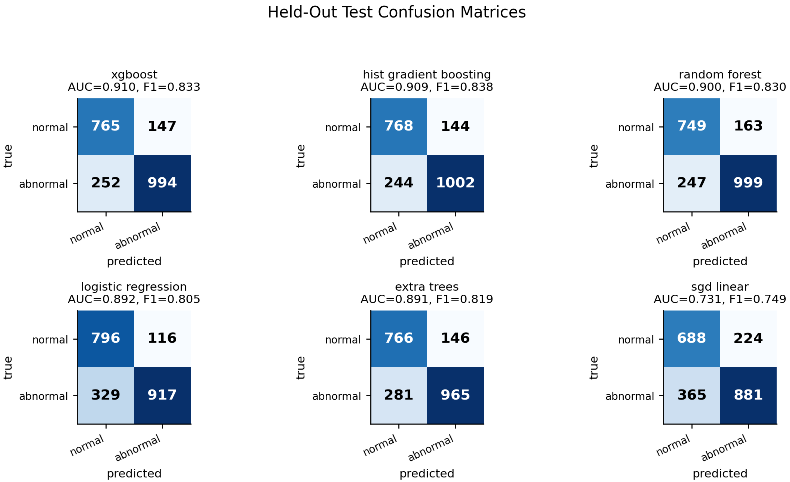
Main comparison

XGBoost obtains the highest test ROC-AUC, while histogram gradient boosting obtains the highest test F1 and balanced accuracy.

Evaluation Visualization



Confusion Matrices



Counts are reconstructed from the reported test precision/recall and official test split class counts: normal=912, abnormal=1,246.

Random Matrix Theory Diagnostic

Marchenko–Pastur reference

For standardized white-noise features with aspect ratio $\gamma = p/n$, sample-covariance eigenvalues concentrate on

$$\lambda_\pm = (1 \pm \sqrt{\gamma})^2.$$

- Eigenvalues inside $[\lambda_-, \lambda_+]$ form the empirical bulk.
- Eigenvalues above λ_+ are structured outliers.
- The experiment checks whether model scores survive if the data are projected onto outlier or bulk subspaces.

PCA Eigenvalue Problem

What PCA solves

For the standardized design matrix $X \in \mathbb{R}^{n \times p}$,

$$S = \frac{1}{n-1} X^\top X = V \Lambda V^\top, \qquad z_{ij} = x_i^\top v_j.$$

PCA orders directions v_j by sample eigenvalues λ_j.

- In low dimensions, large sample variance is often a useful signal heuristic.
- In high dimensions, even pure noise has a spread-out sample spectrum.
- The question becomes: which PCA directions are outside what noise alone would produce?

Marchenko–Pastur as a PCA Cutoff

Noise reference

If standardized features were independent noise and $p/n \to \gamma$, the noise eigenvalues concentrate in

$$\lambda_\pm = (1 \pm \sqrt{\gamma})^2.$$

Observed feature map

- $n = 17{,}084$, $p = 594$.
- $p/n = 0.0348$.
- MP support: $[0.662, 1.408]$.
- 60 eigenvalues exceed λ_+.

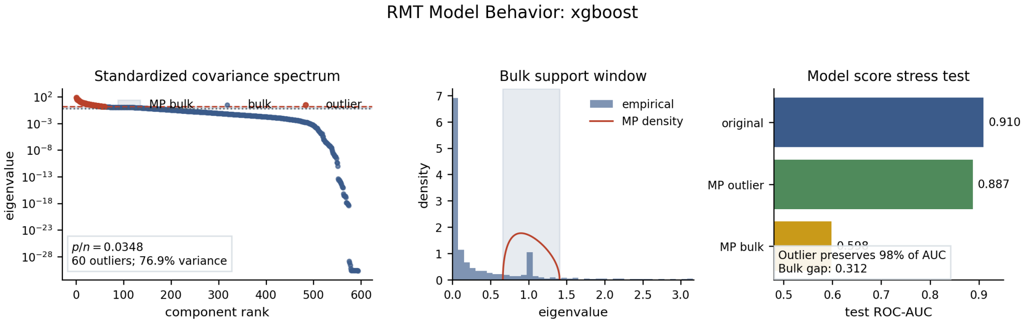
Components above λ_+ are treated as candidate structured directions; components inside the bulk are not automatically useless, only statistically compatible with noise under the reference model.

Projection, Bias, and Variance

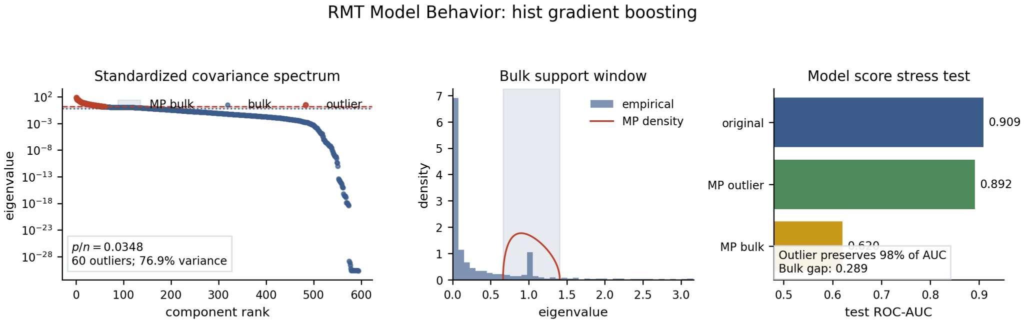
$$I_{\mathrm{out}} = \{j : \lambda_j > \lambda_+\}, \qquad P_{\mathrm{out}} = V_{I_{\mathrm{out}}} V_{I_{\mathrm{out}}}^\top, \qquad P_{\mathrm{bulk}} = I - P_{\mathrm{out}}.$$
$$X_{\mathrm{out}} = XP_{\mathrm{out}}, \qquad X_{\mathrm{bulk}} = XP_{\mathrm{bulk}}.$$

- Hard PCA/RMT filtering keeps X_{out} and discards X_{bulk}.
- This can lower estimator variance by removing unstable directions.
- It can also add approximation bias if weak real signal lives inside the bulk.
- In this run, original XGBoost test ROC-AUC was 0.9097, while MP-outlier reconstruction was 0.8957 and PCA++ scores were 0.8938.

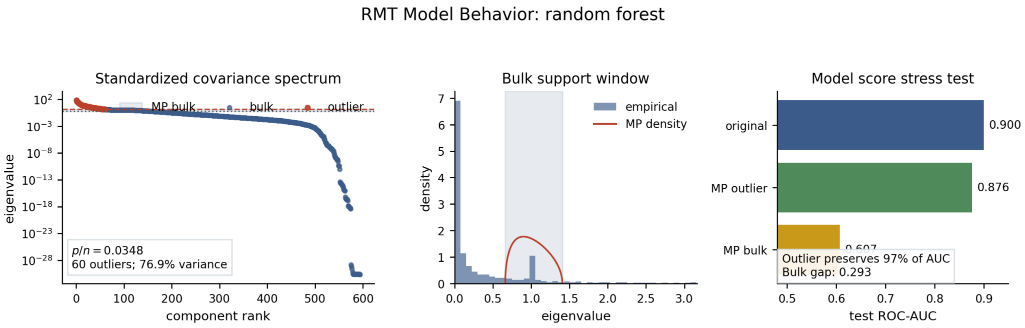
RMT Model Behavior: XGBoost



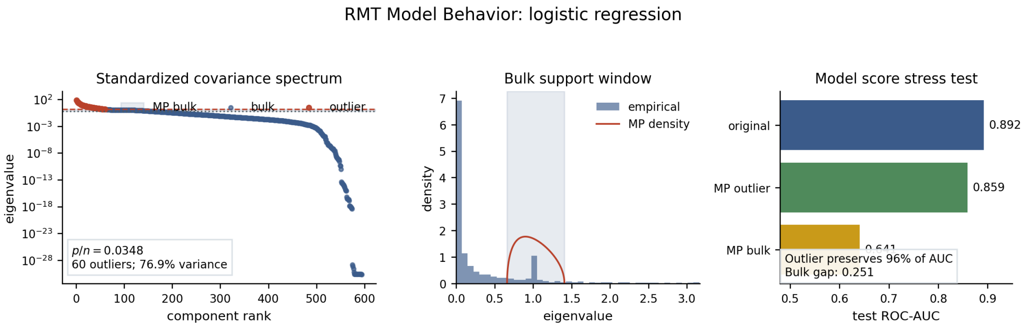
RMT Model Behavior: Histogram Gradient Boosting



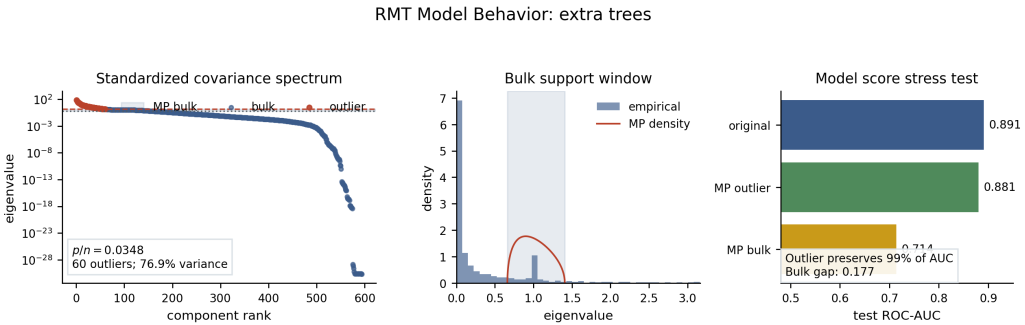
RMT Model Behavior: Random Forest



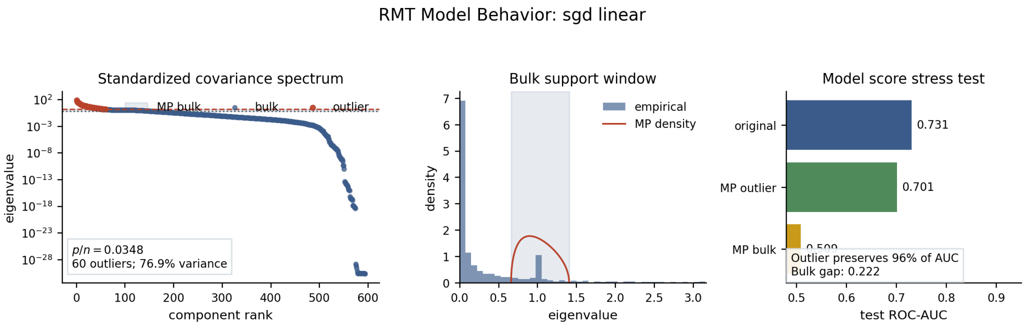
RMT Model Behavior: Logistic Regression



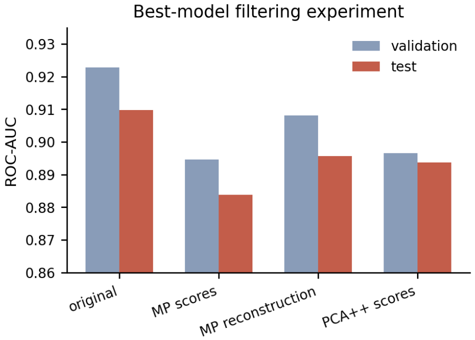
RMT Model Behavior: Extra Trees

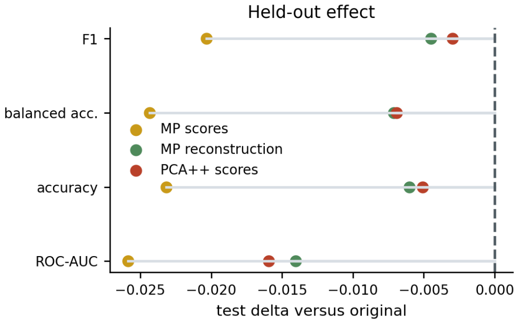


RMT Model Behavior: SGD Linear



Filtering Experiment





Interpretation

- The covariance spectrum has $p/n = 0.0348$, MP support $[0.662, 1.408]$, and 60 outlier components.
- Those outliers explain 76.9% of standardized variance, so the feature map has a compact dominant subspace.
- MP-outlier reconstructions preserve much of the predictive signal.
- MP-bulk reconstructions are weaker but not empty; hard filtering removes weak useful information.
- This explains why RMT is useful as model analysis here, but not as a direct accuracy booster.

Thank you